

Virtuoso[®] Layout Pro: T2 Create and Edit Commands

Course Version IC 25.1

Lecture Manual

Revision 1.0

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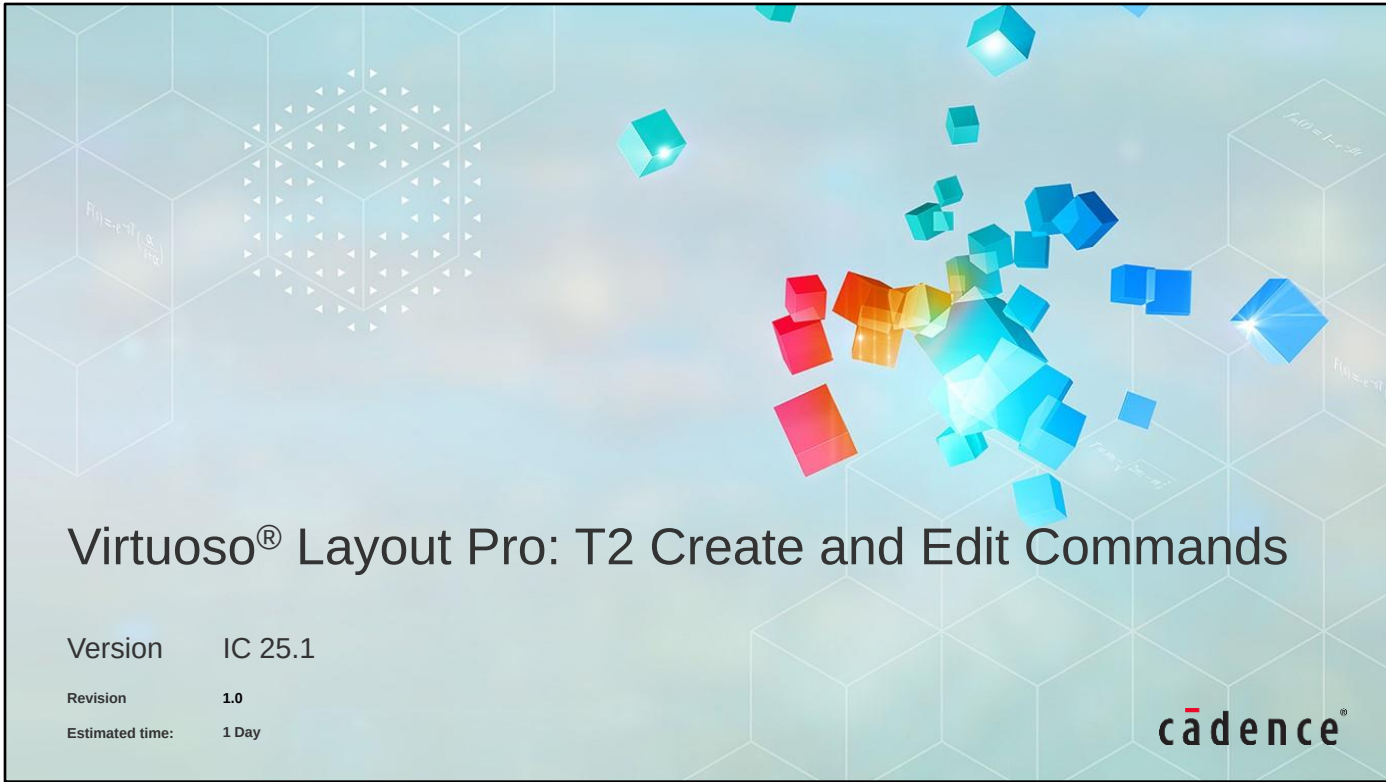
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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following course:

- [Virtuoso Layout Pro: T1 Environment and Basic Commands](#)

Or you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux



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Course Objectives

In this course, you

- Use the features of the Create Via command
- Automatically generate labels with existing information using the Create Label command
- Use the various options of the Copy and Repeat Copy commands
- Generate shapes using Layer Generation Booleans
- Stretch objects using various connected and spine options
- Use the Search and Navigator Assistants to find and identify components
- Save and Restore a selected set of objects to retrieve it later
- Work with Groups to filter and manage design objects
- Find and replace objects
- Remaster instances on your design hierarchy



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Course Agenda

- Create and Edit Commands
 - Using the Create Via Command
 - Using the Create Label Command
 - Using the Copy and Repeat Copy Commands
- Advanced Edit Commands
 - Sizing and Layer Generation
 - Using the Stretch Options
- Hierarchical Editing
 - Using the Search, Save/Restore, and Navigator Assistant
 - Working with Groups
 - Searching the Database
 - Remastering Instances



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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/85088.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.



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How do I register to take the exam?

- Log in to our [Learning Management System](#), click on the course in your transcript, and go to the Content tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- [Credly](#) validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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Icons Used in This Class



Best Practice



Language Syntax



Concept/
Glossary



Frequently Asked
Questions /
Quiz



Error Message

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.



Problem & Solution



Quick Reference



Tool Command



How To



Lab List

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Module 2

Create and Edit Commands

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Module Objectives

In this module, you

- Use the features of the Create Via command
- Automatically generate labels with existing information using the Create Label command
- Use the various options of the Copy and Repeat Copy commands



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Creating Contacts and Vias: Create Via Form



In the layout canvas, choose the **Create – Via** menu command or use the **Create Via** icon from the Create toolbar or press the bindkey **O** to display the Create Via form.



Mode controls the type of via created: **Single**, **Stack**, **Auto**, and **Fast Edit**.

Select the **Mode** required to create the vias.

Create Via Form

Contacts are defined in the technology file. You may place any contact or via individually or use Auto Via. Auto Via places the correct number of via cuts and layer enclosure to fit the area of the intersecting wires.

- **Mode** controls the type of via created. Environment variable: **viaMode**.
 - **Single** creates a single via or an array of vias. Refer to the **Single Mode Options**.
 - **Stack** places the vias belonging to adjacent layers on top of each other. Refer to the **Stack Mode Options**.
 - **Auto** automatically places a via at the intersection of two paths. Refer to the **Auto Mode Options**.
 - **Fast Edit** lets you perform interactive via editing. Refer to the **Fast Edit Mode Options**.
- Note:** In Auto and Fast Edit modes, the Create Via form does not open automatically. To open the Create Via form, press **F3**.
- **Options** opens the **Create Via Options** form.

The Create Via form displays different options based on the mode selected for creating a via:

- Single Mode Options
- Stack Mode Options
- Auto Mode Options
- Fast Edit Mode Options

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Creating Contacts and Vias: Single Mode

Choose **Create – Via** to open the Create Via form and select **Single**.

Single creates a single via or an array of vias.

When off, allows enclosures to be less than minimum.

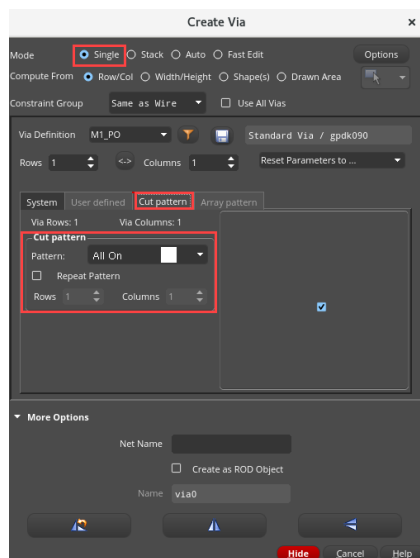
Single Mode Options

The **Via Definition** list displays the valid viaDefs and valid via variants in a tree-like structure from the active constraint group (set in the *Via* list in the Layout Editor Options form). By default, the viaDefs are listed in the same order as specified in the *validVias* constraint. If the *validVias* constraint does not exist in the active constraint group, the viaDefs are ordered by mask number. You can further update the order using the appropriate *Sort* option in the Via Definition Selection widget.

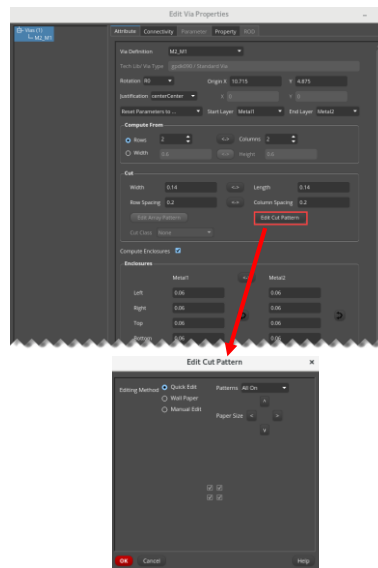
If any filtering options are set up using the *Filter* widget, the *Via Definition* list displays only the viaDefs or via variants that satisfy the filtering options. Each node in the tree represents a standard or custom viaDef. A node with a plus sign (+) represents a parent standard or custom viaDef with child viaVariants. For more information about viaVariants, see *Technology File Via Definitions and Via Specifications* in the *Virtuoso Technology Data ASCII Files Reference, Product Version IC25.1*.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Create Via: Cut Pattern



Placed vias can be edited.



This is how you control via density.

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The Cut Pattern can be edited for already placed vias as well.

Select a via, open the Edit Via Properties form and click **Edit Cut Pattern**.

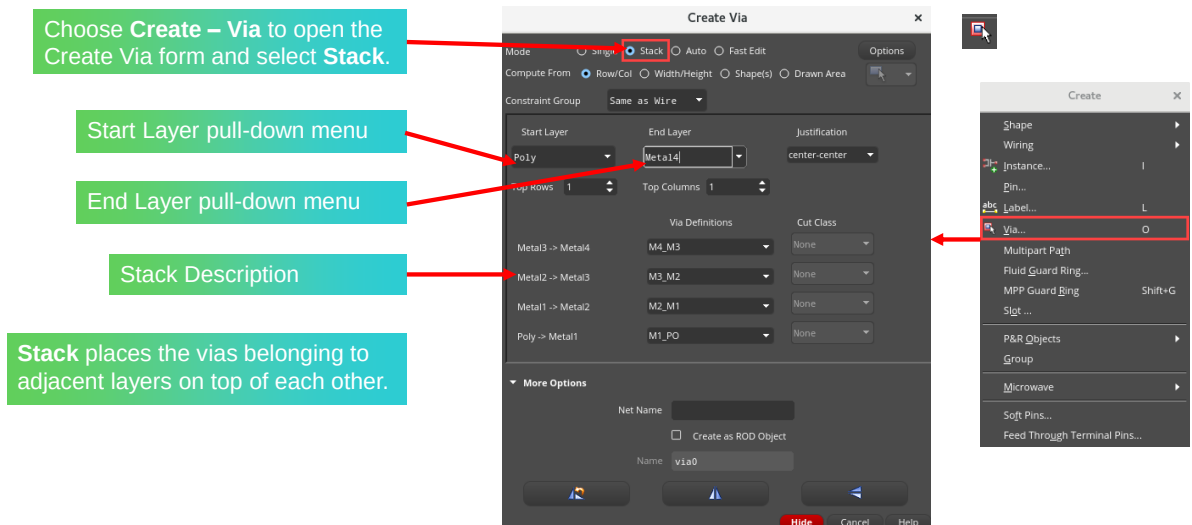
Edit Cut Pattern Form

Note: Edit Cut Pattern is applicable for standard and custom vias.

- **Quick Edit** allows you to edit the cut pattern in conjunction with the *Patterns* cyclic field to select a predetermined pattern.
- **Wall Paper** allows you to edit the cut spaces cut by cut or use the *Patterns* cyclic field to select a predetermined pattern.
- **Paper Size** allows you to add or delete rows or columns to the cut pattern when *Wall Paper* is selected.
- **Manual Edit** displays all of the rows and columns. You can edit the cut spaces cut by cut or use the *Patterns* cyclic field to select a predetermined pattern.
- **Patterns** lets you select a predetermined pattern, such as All On, All Off, Top, Bottom, Left, Right, H Stripes, V Stripes, Chess Board, U-L Corner, U-R Corner, L-L Corner, L-R Corner, Cross, Farm, and As Is.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Creating Contacts and Vias: Stack Mode



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Stack Mode Options

Start Layer lets you select one of the two layers between which you want to place the via stack. The layer selected in the *End Layer* list does not appear in the *Start Layer* list.

End Layer lets you select one of the two layers between which you want to place the via stack. The layer selected in the *Start Layer* list does not appear in the *End Layer* list.

Justification sets the origin of a via. Cyclic field choices are *lowerLeft*, *centerLeft*, *upperLeft*, *lowerCenter*, *centerCenter*, *upperCenter*, *lowerRight*, *centerRight*, *upperRight*, and *offset*.

Top Rows and **Top Columns** set the number of rows and columns for the top via in the via stack. The default number of rows and columns is 1. Rows and columns cannot be specified for the individual vias in the via stack; they apply to the complete via stack. The command calculates the rows and columns of the vias below the top via in such a way that all vias in the stack coincide (environment variables: *viaStackRows* and *viaStackColumns*).

If the via stack contains custom vias, the default rows and columns of the custom vias are used for creating the via stack. You cannot modify the rows and columns of custom vias using this form.

Via Definitions lets you select viaDefs for each layer pair between the specified *Start Layer* and *End Layer*. For each layer pair, the drop-down list displays the available viaDefs. The viaDefs are populated from the *validVias* constraint in the current constraint group in the technology file.

The **Cut Class** cyclic field displays cut class choices only if a *cutClasses* constraint has been defined for the cut layer of the selected viaDef in the *layerRules* section of the technology file. If the *cutclassGroups* are defined in the technology file, you can choose a cut class group for all the layer pairs.

Creating Contacts and Vias: Auto Mode

Choose **Create – Via** to open the Create Via form and select **Auto**.

The size and number of cuts will fill the area where the shapes overlap.

Connects the shapes that are on the same net.

Specifies the preferred Orientation for the Cut BBox as either **auto**, **horizontal**, or **vertical**.

Auto automatically places a via at the intersection of two paths.

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Auto Mode Options

When you select this option, the form displays the following additional fields:

- **Use Entire Overlap** uses the entire overlap to create the via. If this option is not selected, an intersection between the user area and the overlap box is used to create the via. Environment variable: *viaUseEntireOverlap*
- **Cut Class** is a cyclic field that displays cut class choices only if a *cutClasses* constraint has been defined for the cut layer of the selected viaDef in the *layerRules* section of the technology file.
- **Via Alignment** specifies the alignment of the via using the alignment choices: *auto*, *top-left*, *top-center*, *top-right*, *center-left*, *center-center*, *center-right*, *bottom-left*, *bottom-center*, and *bottom-right*. The default is *auto*. Environment variable: *viaAutoViaAlignment*
- **Cut Class Orientation** specifies the preferred orientation for the cut class as one of the following: *auto*, *horizontal*, *vertical*, or *uniform in stack*. Environment variable: *viaCutClassOrientation*
- **Cut BBox Orientation** specifies the preferred direction for the cut box as one of the following: *auto*, *horizontal*, or *vertical*. Environment variable: *viaAutoViaCutArrayDirection*
- **Min Number of Cuts** specifies the minimum number of cuts to be automatically created for any overlapping area between two shapes. Environment variable: *viaAutoViaMinNumCuts*
- **Max Stack Depth**, when selected, the overlap with the same number of transitions or less than the specified number are searched. You can specify the value of the maximum stack depth. If you specify 1 as the value in the *Max Stack Depth* field, and the layers under *Metal11* are *Metal2* and *Metal5*, only the overlap *Metal1-Metal2* is retained. The other overlaps, *Metal1-Metal5* (stack depth=4) and *Metal2-Metal5* (stack depth=3) are dropped. Environment variable: *viaAutoViaEnableMaxStackDepth* and *viaAutoViaMaxStackDepth*
- **Connect Shapes On The Same Net** connects the shapes that are on the same nets.

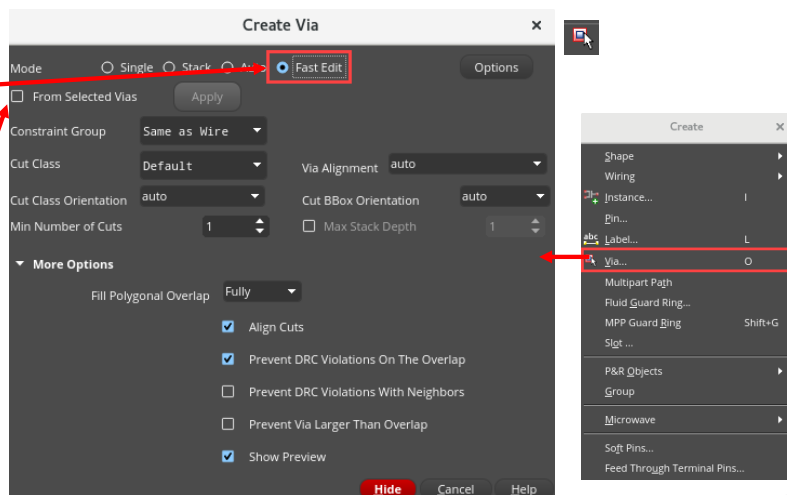
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Creating Contacts and Vias: Fast Edit Mode

Choose **Create – Via** to open the Create Via form and select **Fast Edit**.

Use **From Selected Vias** to fast edit from a set of selected vias.

Use the Fast Edit via mode to perform interactive via editing.



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Fast Edit Mode Options

Use the Fast Edit via mode if you want to perform interactive via editing.

From Selected Vias Lets you use fast edit command from a set of selected vias. Environment variable: **viaFastEditFromSelectedVias**

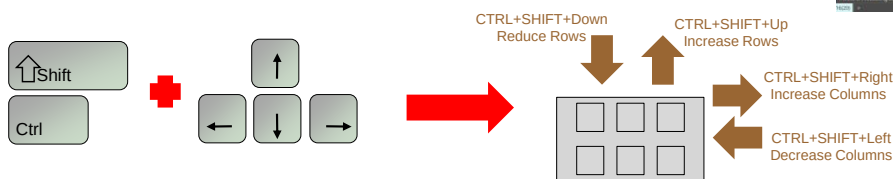
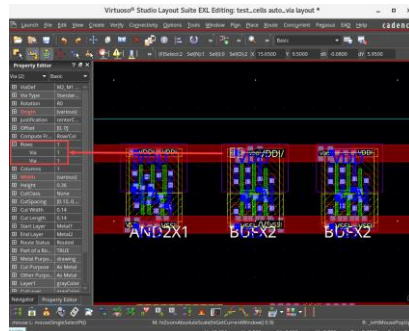
The other options available in the Fast Edit via mode are same as the Auto via mode options.

Adjusting the Placed Vias Using the Property Editor Assistant



In the layout canvas, choose the *workspace* as **Basic** to display the Property Editor assistant.

- The number of Rows/Columns of an already placed via can be easily increased using the Property Editor assistant.
- To provide a more user-friendly method, a set of bindkeys has been defined that lets you interactively modify the dimensions of the placed Via.



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Property Editor Assistant

If multiple objects are selected, the cyclic field in the top-left corner of the Property Editor assistant can be used to view only a specific category (Via).

To change the size of a given via, the corresponding parameter should be expanded by clicking the *plus* symbol (in the example shown here, the *Rows* parameter has been expanded), and then the *line* corresponding to the targeted via should be edited. The via which you are editing is highlighted. In the example on this page, a blue frame around the edited via is visible.

When the edit is finished, a click out of the edited field is required to make the changes.

Customized Bindkeys

The bindkeys described here rely on SKILL® code which is included in the Cadence® Learning and Support Article ID 11777642.

Refer to Article#11777642 from Cadence Learning and Support from this link:

[How to increase or decrease rows and columns in a via, increase or decrease number of its cuts](#)

This SKILL code should be adapted to reflect the specialties of the technology you are using.

Note: If a bindkey is already in use by the window manager you are using, such as GNOME or KDE, this bindkey will not be available inside the Virtuoso® platform.

Invoking the Create Label Form



In the layout canvas, choose the **Create – Label** menu command or use the **Create Label** icon from the Create toolbar or press the bindkey **L** to display the Create Label form.



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Create Label Form

- **Mode** sets the method for generating labels.
- **Manual** derives the label name from the text string specified in the *Label (Pattern)* field and places the label at the point of click. *Manual* is the default mode of creating labels. You can place multiple same or different labels on a single object. In the *Manual* mode, you can label any type of shape and object except the DB object types “textDisplay” and “label”. You can also add labels in empty spaces.
- **Auto** automatically generates labels from the connectivity information on objects and places the label at the point of click. *Auto* also automatically generates labels for instances from the instance name or instance master cell and places the label at the center of the instance.
 - Two tabs become available in this mode: From Interconnect and From Instance. There is a checkbox above each tab. You can select one or both the checkboxes to enable label creation from connectivity information or instances or both.
 - In the *Auto – From Interconnect* mode, if the shape to be labeled is a pin, the label name is derived from the pin’s terminal name. For other shapes, the label name is derived from the net name associated with the shape. You can place multiple labels on a single object, but all the labels on an object will have the same label name, derived from the object’s connectivity information.
- **Label (Pattern)** specifies the text or indexed pattern to use for assigning label names. You specify the pattern inside the < and > symbols. These symbols are retained in the generated labels and represent bus bits. You can also use the [and] symbols to enclose the pattern, in which case, the symbols are not part of the labels generated.
 - The *Label (Pattern)* field is active only in the *Manual* mode. In the *Auto* mode, the label names are generated automatically from connectivity information of shapes or from instances. Therefore, the *Label (Pattern)* field is not available in the *Auto* mode.
 - The form comprises the following sections: *Label Layer/Purpose*, *Label Options*, *Scan Line*, and *Auto Step*.
 - These sections and various options in these sections are available based on the mode you select to create labels.
 - The following tabs are available in the *Auto* mode: From Interconnect and From Instance.

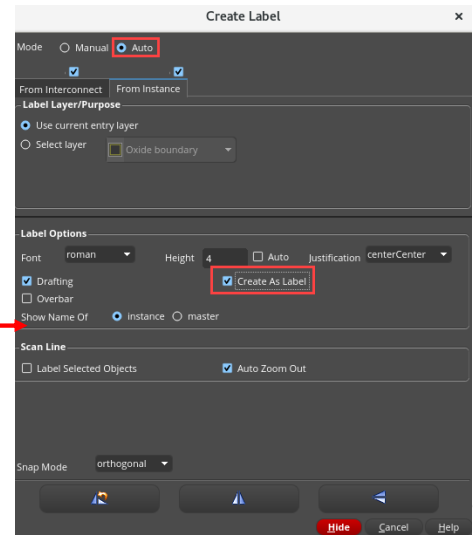
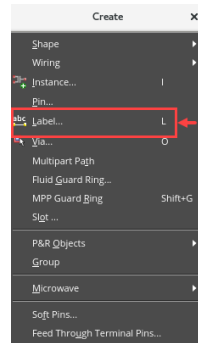
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

“Create As Label” Option in the Create Label Form



In the layout canvas, choose the **Create – Label** menu command or use the **Create Label** icon from the Create toolbar or press the bindkey **L** to display the Create Label form. In the **Auto** mode, choose the **Create As Label** option to create the label as “*Label*” instead of “*TextDisplay*”.

- In earlier versions, in the **Auto** mode of the Create Label command, only “*TextDisplay*” could be created.
- From IC 6.1.7, the user can choose to create *TextDisplay* or *Label*. When the **Create As Label** option is selected, “*Label*” will be created instead of “*TextDisplay*”.
- It is useful when labels need to be modified to create virtual connects. For example, change the label “VDD” to “VDD:”; the net name and the label name are different.
- Use **layout** envvar **createAsLabel** to change the default value of this option.



From Interconnect

In this *Auto* mode, the form comprises the following sections:

Label Layer/Purpose: All the options are available in this mode.

Label Options: *Font*, *Height*, *Justification*, *Drafting*, *Create As Label*, and *Overbar* are available in this mode.

Create As Label determines whether the object is created as a label or text display. If this option is selected, a label is created, else a text display is created.

Note: When a shape with a textDisplay is removed from a net, the textDisplay is automatically removed as well. If the shape is added back to a net, the textDisplay is also automatically recreated. Environment variable: `createAsLabel`

Scan Line: Both the options are available in this mode.

From Instance

In this *Auto* mode, the form comprises the following sections:

Label Layer/Purpose: *Use current entry layer*, and *Select layer* options are available in this mode.

Label Options: *Font*, *Height*, *Justification*, *Drafting*, *Overbar*, and *Create As Label* checkboxes are available in this mode.

Show Name Of enables you to specify whether to use the *instance* name or the instance *master* cell name for generating the labels.

Scan Line: Both the options are available in this mode.

Snap Mode locks the direction in which you can draw the scan line. You can set it to *anyAngle*, *diagonal*, *orthogonal* (default), *horizontal*, or *vertical*. The default selection is *orthogonal*. The setting in this list is applicable while using the *Scan Line* and *Auto Step (Custom)* methods for generating labels.

Rotate turns the label 90 degrees counterclockwise.

Sideways mirrors the label along the X axis. *Drafting* must be off for this button to work.

Upside Down mirrors the label along the Y axis. *Drafting* must be off for this button to work.

Invoking the Create Pin Form

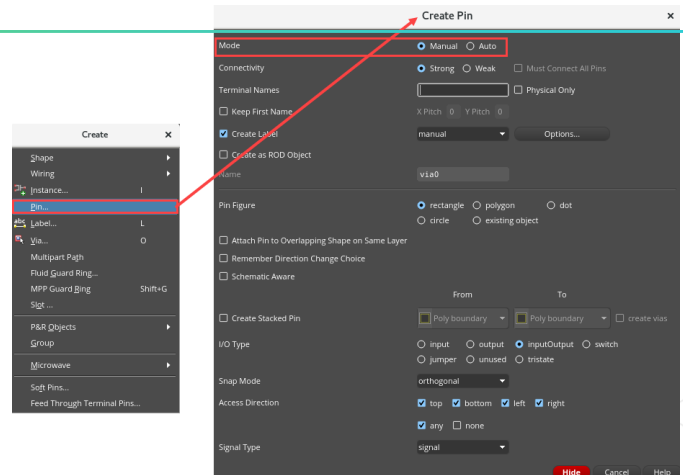


In the layout canvas, choose **Create – Pin** to display the Create Pin form.

Use the **Create Pin** form to create and place the pins.

The form supports the *Manual* and *Auto* modes of pin creation.

- **Manual Mode** – Enables you to create and place the pins manually.
- **Auto Mode** – Enables automatic pin placement.



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Create Pin Form

Use the Create Pin form to create and place pins. The form supports the following modes of pin creation:

Manual Mode Enables you to create and place pins manually.

Auto Mode Enables automatic pin placement.

Environment variable: **pinMode**

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

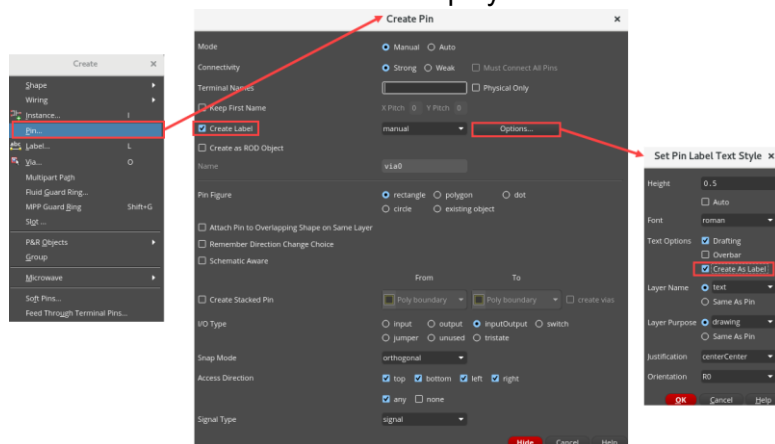
“Create As Label” Option in the Create Pin Form



In the layout canvas, choose **Create – Pin** to display the Create Pin form.

Choose **Create As Label** option to create the labels as “Label” instead of “TextDisplay”.

The **Create As Label** option is added in the “Create Label” **Options** in the Create Pin form. When this option is **ON**, “Label” will be created instead of “TextDisplay”.



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Creating Pins

The Create Pin command lets you create pins in a cellview to connect the cellview to another cell in your design hierarchy. The Create Pin command also lets you define the connectivity of the pin, strong or weak, even while creating the pin. For defining or editing the connectivity of existing pins in the design, you can use the Connectivity – Pins command. Using the Connectivity – Pins command, you can define connectivity as must-join connect, strong connect, weak connect, or pseudo parallel connect. For more information about these connectivity types, see *Overview of Pin Connectivity*.

Create As Label determines whether the pin is created as a label or textdisplay. If this option is selected, a label is created, else a textdisplay is created. Environment variable: `pinTextCreateAsLabel`

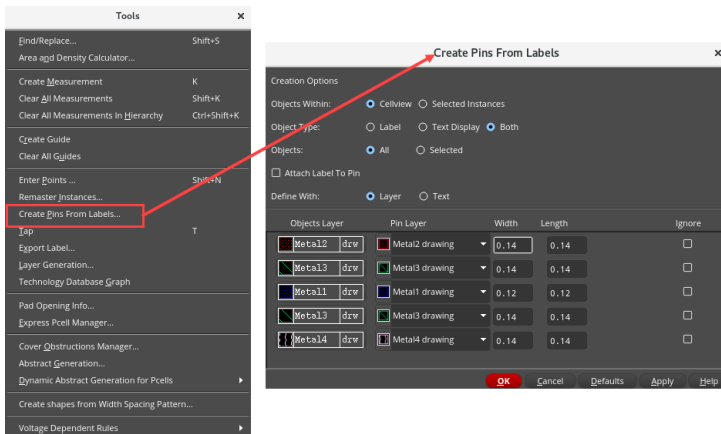
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Using the “Create Pins From Labels” Form



In the layout canvas, choose **Tools – Create Pins From Labels** to display the Create Pins From Labels form.

Use the **Create Pins From Labels** command to create pins from the labels and text displays in your layout cellview.



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Creating Pins From Labels

The *Tools – Create Pins From Labels* command lets you create pins from the labels and text displays in your layout cellview. You can specify the pin dimensions and the layer on which each pin is created. The generated pins are centered on the origin of the label or text display from which they are derived and have terminal names matching those on the corresponding labels or text displays.

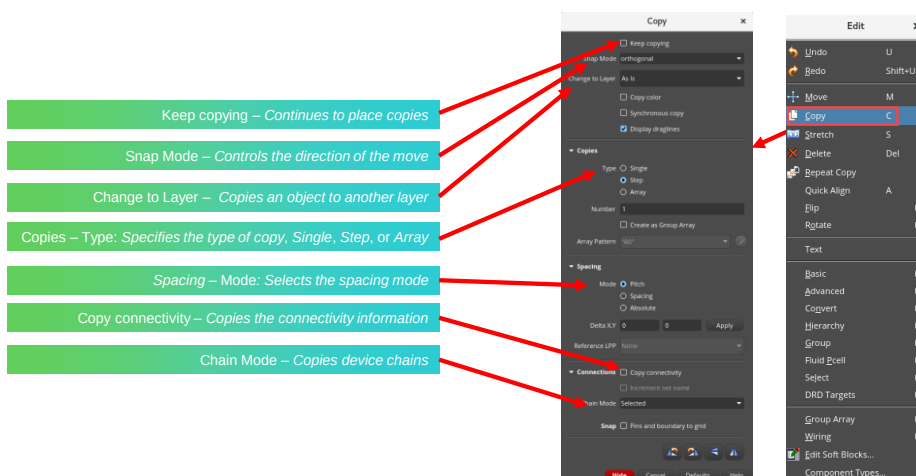
Note: If the text label contains a colon (:), the colon and trailing text are dropped from the name of the generated pin. For example, given a text label VDD:P,28, the resulting pin name is VDD.

If a pin already exists for a selected label or text display, no new label is created for that object. The pin and object are not attached to each other. If you move the pin, the corresponding label or text display does not follow.

Invoking the Copy Form



In the layout canvas, choose the **Edit – Copy** menu command or use the **Copy** icon from the Edit toolbar or press the bindkey **C** to display the Copy form.



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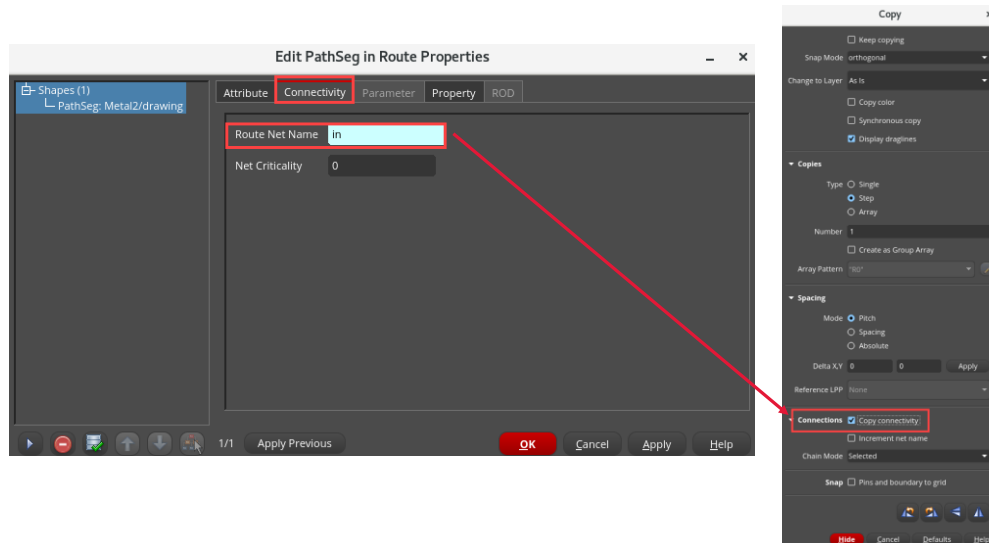
Copy Form

- **Keep Copying** enables you to create multiple copies of an object. With this checkbox selected, a copy of the selected object is created every time you click in the design display area. You can customize this mode of copying by configuring the other settings in the form. The *Delta*, *Spacing*, and *Exact Overlap* checkboxes are not available when you select this checkbox.
- **Snap Mode** controls the direction in which you can move the copied object.
- **Change to Layer** enables you to copy an object to another layer. You can search for the specific layer-purpose pair using the search filter in the *Change To Layer* list box. Environment variable: *copyToLayer*
- **Copies: Type** specifies the type of copy, either *Single*, *Step*, or *Array*.
- **Single**: Lets you create a single copy of objects.
- **Step**: Lets you create a line of copied objects.
- **Array** lets you create an array of copied objects. Use the *Rows* and *Columns* fields to specify the dimensions of the array. The array of copied objects can be placed only in the original cellview; it cannot be copied to another cellview.
- **Number** lets you specify the number of copies to be made.
- **Spacing: Mode** lets you select the spacing mode.
- **Pitch** lets you specify the delta that applies between the objects in an array and the distance is specified in the *Delta X, Y* field. The distance is measured between the same-edge corners of the objects in the array.
- **Spacing** applies between the objects in an array the distance specified in the *Spacing X, Y* fields.
- **Absolute** enables you to copy an object at the location specified in the *X, Y* fields.
- **Delta X,Y** enables you to specify offset for copy.
- **Apply** copies the selected object based on the settings specified in the form.
- **Connections: Copy connectivity** copies the connectivity information on path, pathseg, fluid guard ring, and via to the new object. Environment variable: *copyConnectivity*
- **Chain Mode** enables you to copy device chains.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

“Copy connectivity” Option in the Copy Form

When the *Copy connectivity* option in the **Copy** command is selected, the connectivity information on the path, pathseg, fluid guard ring, and via will be copied to the new object.



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Copy connectivity option in the **Copy** command copies the connectivity information on path, pathseg, fluid guard ring, and via to the new object. Environment variable: *copyConnectivity*

Copying the Objects

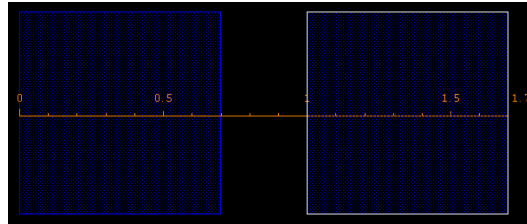


The **Copy** command enables you to place a copy of an object in a cellview.

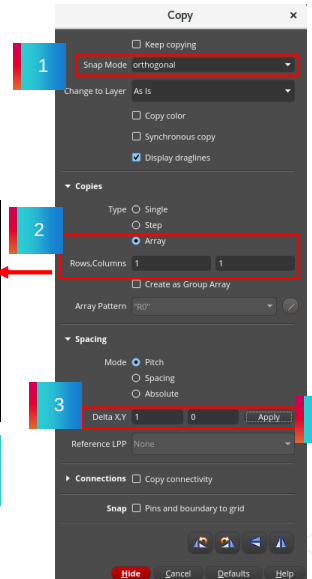
You can copy an object to the same cellview or to another cellview.

In the **Copy** form, select the options as follows:

1. Snap Mode: orthogonal
2. Array: Rows: 1; Columns: 1
3. Delta: X: 1; Y: 0
4. Click **Apply** to make the copy



When you copy an object to another cellview, the constraints associated with an object are not copied to the resultant cellview.



■ In the Copy form, select the options, as follows:

- Snap Mode: orthogonal
- Array: Rows: 1; Columns: 1
- Delta: X: 1; Y: 0
- Click **Apply** to make the copy

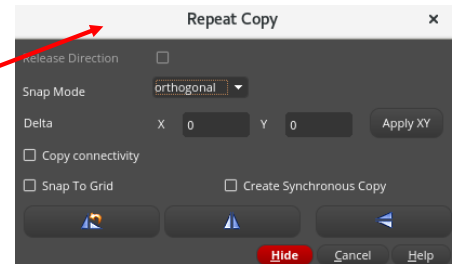
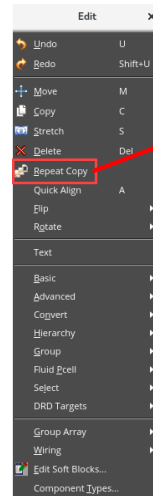
Invoking the Repeat Copy Form



In the layout canvas, choose the **Edit – Repeat Copy** menu command or use the **Repeat Copy** icon from the Edit toolbar to display the Repeat Copy form.



- Select a shape to copy.
- Place the copy manually or use the coordinates.
- Press the Period (. dot) key on the keyboard to continue to step and repeat shapes.



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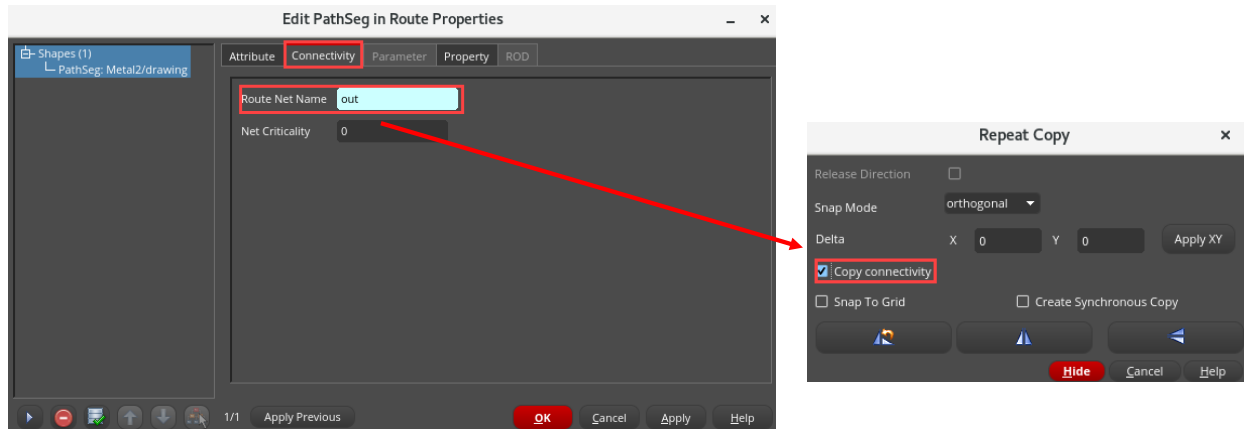


Repeat Copy Form

- **Release Direction** resets the paste direction and allows you to paste the copied object in any direction and at any distance with respect to the previous pasted object. When off, the direction of pasting the object is fixed, which is the direction in which the copied object was last pasted. The checkbox is available only after you have placed the first copy.
- **Snap Mode** controls how the mouse pointer snaps to the grid as you place the copies.
- **anyAngle** enables you to move the pointer at any angle.
- **diagonal** enables you to move the pointer parallel to the X or Y axis or at a 45-degree angle to the axes.
- **orthogonal** enables you to move the pointer parallel to the X and Y axes.
- **horizontal** enables you to move the pointer parallel to the X axis.
- **vertical** enables you to move the pointer parallel to the Y axis.
- **Delta: X and Y** enables you to specify the exact distance from the object at which you want to place the copy.
- **Apply XY** places the copy at the specified distance from the object.
- **Copy Connectivity** copies the connectivity information on path, pathseg, fluid guard ring, and via to the new object. Environment variable: *copyConnectivity*
- **Rotate** turns the object 90 degrees counterclockwise each time it is pressed.
- **Sideways** mirrors the object along the X axis.
- **Upside Down** mirrors the object along the Y axis.

“*Copy connectivity*” Option in the Repeat Copy Form

When the *Copy connectivity* option in the **Repeat Copy** command is selected, the connectivity information on the path, pathseg, fluid guard ring, and via will be copied to the new object.



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Copy Connectivity option in the **Repeat Copy** command copies the connectivity information on path, pathseg, fluid guard ring, and via to the new object. Environment variable: *copyConnectivity*

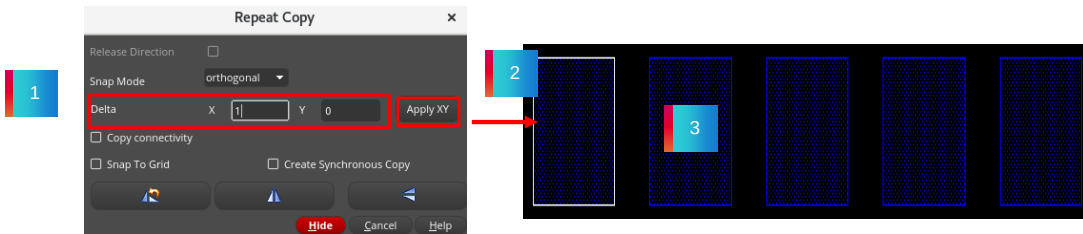
Copying the Objects in Repeat Mode



The **Repeat Copy** command enables you to repeatedly place copies of the selected object. The command supports both pre-selection and post-selection modes.

In the **Repeat Copy** form, select the options as follows:

1. In the Delta field, change the X value to 1.
2. Click **Apply XY** to make the copy. The copy is made once.
3. Press the **Period (dot)** key on your keyboard to continue making copies.



For a simple copy-paste operation, use the **Copy** command.

In the Repeat Copy form, select the options as follows:

- In the Delta field, change the X value to 1.
- Click **Apply XY** to make the copy. The copy is made once.
- Press the period (., dot) key on your keyboard to continue making copies.

Repeat Copy, Move, or Stretch API: `leRepeatCopyMoveStretch()`



- The `leRepeatCopyMoveStretch()` API enables you to repeatedly copy, move, or stretch the selected objects. The command repeats the last run command among the Copy, Move, and Stretch commands in the preselect mode. The command uses the dx and dy from the last copy, move, or stretch operation.
- If the **Move** command was the last run command among the three commands, `leRepeatCopyMoveStretch()` will move the selected object by the same dx and dy.
- If the last command was **Copy**, `leRepeatCopyMoveStretch()` will copy the selected object and place the copy at the same distance (dx and dy) from the original.

For example:

- Select object **A**.
- Invoke the **Copy** command and make a copy of **A**. Keep the new copy selected.
- Invoke `leRepeatCopyMoveStretch()`, and another copy will be copied and placed with the same spacing (dx and dy).
- Invoke `leRepeatCopyMoveStretch()` again to make another copy.
- You can assign a bindkey for the `leRepeatCopyMoveStretch()` API.

▪ For example:

```
hiSetBindKey( "Layout" "Shift Ctrl<Btn2Down>(2)" "leRepeatCopyMoveStretch()")
```



You can use the `leRepeatCopyMoveStretch()` SKILL function to copy, move, or stretch the selected objects and apply the same transform as done in the last copy, move, or stretch command and select the newly created object.

Quiz: The Modes Available in the Create Via Form



You can use the **Create Via** form to place contact/via in the following modes (select all that apply):

- A. Single
- B. Multiple
- C. Stack
- D. Auto
- E. Fast Edit
- F. All of the above

Answers: A, C, D, and E

You can use the **Create Via** form to place contact/via in the *Single*, *Stack*, *Auto*, and *Fast Edit* modes.



Answers: A, C, D, and E

You can use the **Create Via** form to place contact/via in the *Single*, *Stack*, *Auto*, and *Fast Edit* modes.

Quiz: Changing the Rows/Columns of a Placed Via



The number of **Rows/Columns** of an already placed via can be easily changed using (select the correct choice):

- A. World View Assistant
- B. Annotation Browser Assistant
- C. Property Editor Assistant
- D. Palette Assistant
- E. None of the above

Answer: C

The number of **Rows/Columns** of an already placed via can be easily changed using the *Property Editor* Assistant.



Answer: C

The number of **Rows/Columns** of an already placed via can be easily changed using the *Property Editor* Assistant.

Quiz: The Label (Pattern) Field in the Create Label Form



In the **Create Label** form, the *Label (Pattern)* field is active in both the *Manual* and the *Auto* modes.

- A. True
- B. False

Answer: B

In the **Create Label** form, the *Label (Pattern)* field is active only in the *Manual* mode. In *Auto* mode, the label names are generated automatically from the connectivity information of shapes or instances. Therefore, the *Label (Pattern)* field is not available in the *Auto* mode.



Answer: B

In the **Create Label** form, the *Label (Pattern)* field is active only in the *Manual* mode. In the *Auto* mode, the label names are generated automatically from connectivity information of shapes or from instances. Therefore, the *Label (Pattern)* field is not available in the *Auto* mode.

Quiz: Making More Copies Using the Repeat Copy Command



Name the key you can use on the keyboard to continue making more copies with the **Repeat Copy** command. ----- Enter the answer in the blank field.

Answer: Period (dot) key

You can use the Period (dot) key on the keyboard to continue to step and repeat shapes with the **Repeat Copy** command.



Answer: Period (dot) key

You can use the Period (dot) key on the keyboard to continue to step and repeat shapes with the **Repeat Copy** command.

Quiz: The SKILL Function to Repeatedly Copy, Move, or Stretch



What is the SKILL® function used to repeatedly copy, move, or stretch the selected objects?
----- Enter the answer in the blank field.

Answer: *leRepeatCopyMoveStretch()* SKILL function

You can use the *leRepeatCopyMoveStretch()* SKILL function to repeatedly copy, move, or stretch the selected objects.



Answer: *leRepeatCopyMoveStretch()* SKILL function

You can use the *leRepeatCopyMoveStretch()* SKILL function to repeatedly copy, move, or stretch the selected objects.

Quiz: The Bindkey of the Create Via Command



What is the bindkey used to invoke the **Create Via** command?

You can use the bindkey **O** to invoke the **Create Via** command.



What is the bindkey used to invoke the **Create Via** command?

You can use the bindkey **O** to invoke the **Create Via** command.

Quiz: The Modes Available in the Create Label Form



What are the 2 modes available in the **Create Label** form?

You can create label either in **Manual** mode or in **Auto** mode.



What are the 2 modes available in the **Create Label** form?

You can create label either in **Manual** mode or in **Auto** mode.

Quiz: The Bindkey of the Create Label Command



What is the bindkey used to invoke the **Create Label** command?

You can use the bindkey **L** to invoke the **Create Label** command.



What is the bindkey used to invoke the **Create Label** command?

You can use the bindkey **L** to invoke the **Create Label** command.

Quiz: Creating Label Using the Create Label Form



In the **Create Label** form in the **Auto** mode section, which option should be selected so that “*Label*” will be created instead of “*TextDisplay*”?

In the **Create Label** form in the **Auto** mode section, when the “**Create As Label**” option is selected, “*Label*” will be created instead of “*TextDisplay*”.



In the **Create Label** form in the **Auto** mode section, which option should be selected so that “*Label*” will be created instead of “*TextDisplay*”?

In the **Create Label** form in the **Auto** mode section, when the “**Create As Label**” option is selected, “*Label*” will be created instead of “*TextDisplay*”.

Quiz: The “*Copy connectivity*” Option



Name the commands where the *Copy connectivity* option is available?

The *Copy connectivity* option is available in the **Copy** and **Repeat Copy** commands.



Name the commands where the *Copy connectivity* option is available?

The *Copy connectivity* option is available in the **Copy** and **Repeat Copy** commands.

Module Summary

In this module, you learned how to

- Use the features of the Create Via command
- Automatically generate labels with existing information using the Create Label command
- Use the various options of the Copy and Repeat Copy commands



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Labs



Lab 2-1 Using the Create Via Command

Lab 2-2 Using the Create Label Command

Lab 2-3 Using the Copy and Repeat Copy Commands



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Module 3

Advanced Edit Commands

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Module Objectives

In this module, you

- Generate shapes using Layer Generation Booleans
- Stretch objects using various connected and spine options



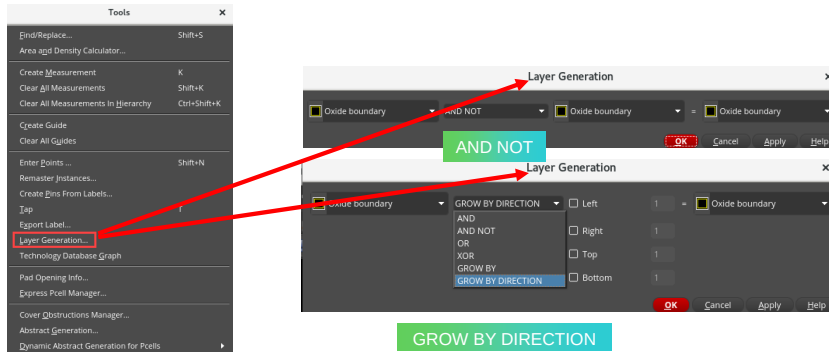
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Invoking the Layer Generation Form



In the layout canvas, choose the **Tools – Layer Generation** menu command to invoke the Layer Generation form.

- Use logical functions to create shapes.
- Boolean operators available are AND, AND NOT, OR, XOR, GROW BY, and GROW BY DIRECTION.



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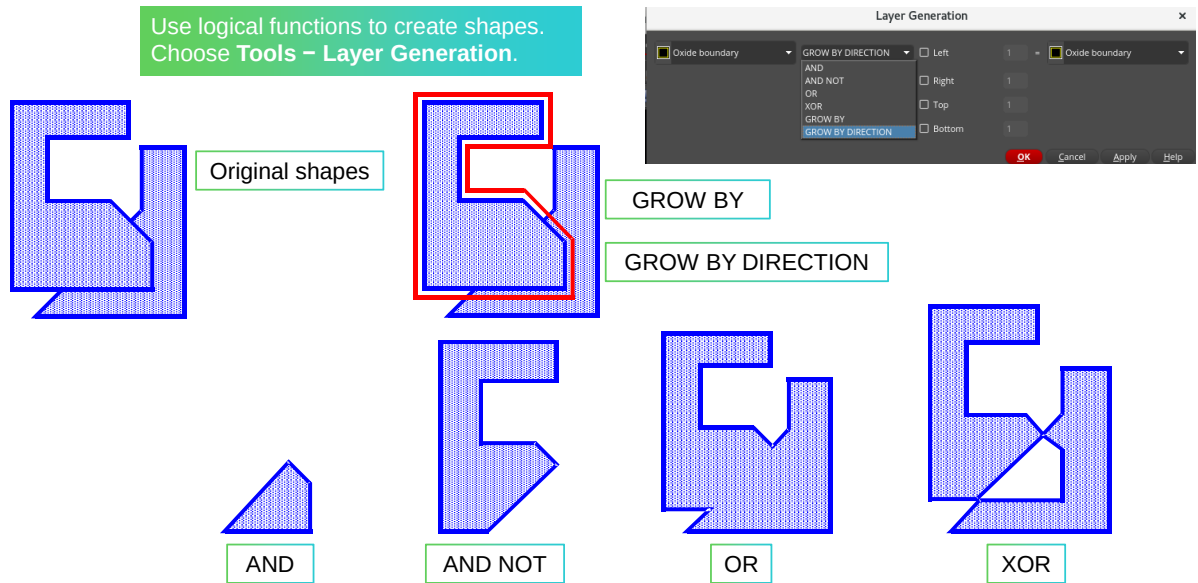


Layer Generation Form

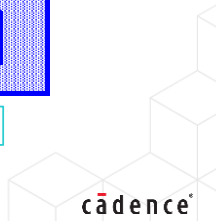
- **Input Layers** (the first and third fields) sets the layer or layers you want to modify with *Layer Generation*. You can search for specific layer-purpose pair using the search filter in the first and third fields.
- **Boolean Operators** (second field) combines the selected objects on the input layers.
- **GROW BY** redraws the Layer Generation form so you can enter a resizing value. Enter the number of user units (resizing value) by which to enlarge selected objects on the input layer in the field to the right.
- **Output Layer** (fourth field) sets the layer on which the new shapes will appear. You can search for specific layer-purpose pair using the search filter.

Note: The layer selection using the search filter is available for the *Input Layers* and *Output Layer* fields only if the *Use Layer Palette LPPs Only* option is selected in the Layout Editor Options form.

Using the Layer Generation Boolean Functions



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Boolean Functions

The Layer Generation command lets you use Boolean functions to create a new layer. You can choose from the six different functions, namely *AND*, *AND NOT*, *OR*, *XOR*, *GROW BY*, and *GROW BY DIRECTION*, on the Layer Generation form. The system uses the first layer and performs the logical function with the second layer, creating a new object on the third layer.

Sizing Function

To create a new object with a sizing operation, use the *GROW BY* and *GROW BY DIRECTION* options. These options are similar to the **Size** command, except you specify the layer on which the sized object is created. A nonpolygonal shape is converted to a polygon.

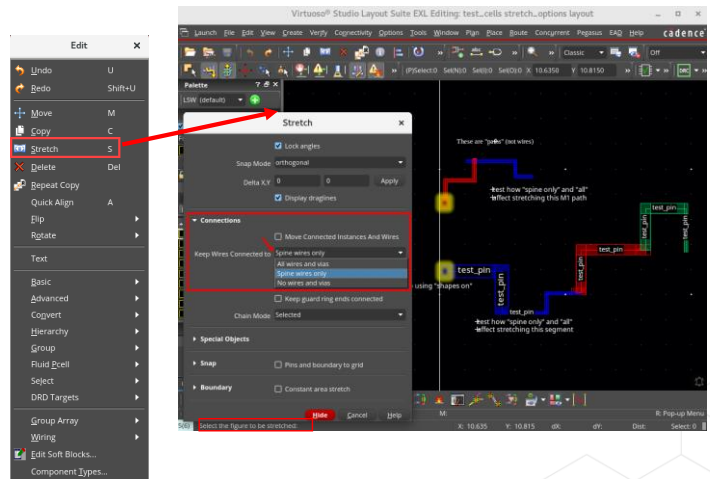
Invoking the Stretch Form



In the layout canvas, choose the **Edit – Stretch** menu command or use the Stretch icon from the Edit toolbar or press the bindkey S to open the Stretch form.

- Use the Stretch form to stretch objects in the specified window.
- Keep Wires Connected to: Stretches paths, pathsegs, and via vertices coincident with a selected vertex. These are stretched as a set.
 - All wires and vias: Maintains pathseg-to-pathseg and pathseg-to-via connections during stretch.
 - Spine wires only: Maintains connections to spine pathsegs but not to connecting vias when stretching a spine. This option is selected by default.
 - No wires and vias: Does not maintain pathseg-to-pathseg and pathseg-to-via connections during stretch.

Stretch with Keep Wires Connected to: All wires and vias
 Stretch with Keep Wires Connected to: Spine wires only (default)
 Stretch with Keep Wires Connected to: No wires and vias



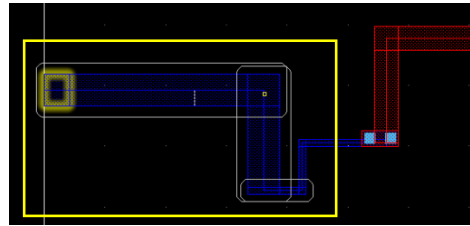
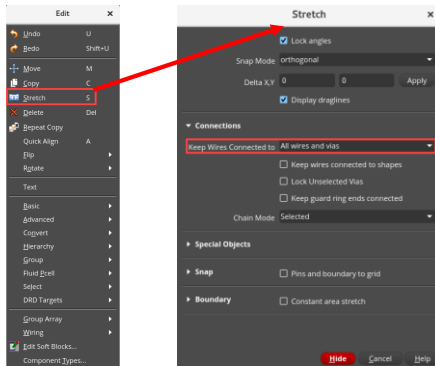
Invoking the Stretch Form

Use the **Stretch** form to stretch objects in the specified window.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Stretch: Keep Connected: All Wires and Vias

Example of **Keep Wires Connected to**: Choose **All wires and vias**.



All connected wires and vias will stretch.



Use the **Stretch** command to stretch diagonal edges, intersecting wires, and pathSegs or paths by changing the vertices of the XY locations. When an isolated fully selected via, pathSeg, or path is stretched, the object is moved to a new location similar to using the Move command. The *Keep Wires Connected to* option on the Stretch form allows stretched wires to remain connected.

Note: PathSegs are supported by advanced features. Advanced features such as *Push* are activated when using pathSegs. Path objects are not supported by advanced features.

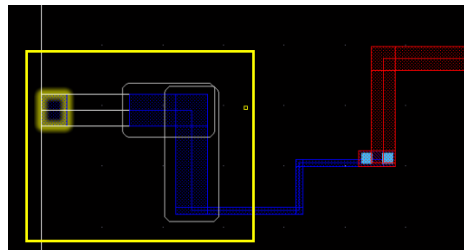
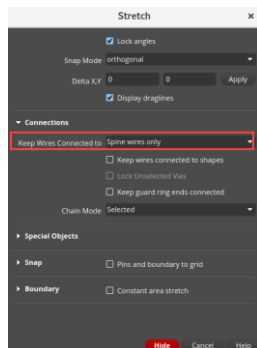
The Stretch command does not keep overlapping wires connected if they are on different nets. In Virtuoso® Layout Suite XL, overlapping wires on different nets are kept connected when stretched if their endpoints meet exactly.

When a wire connects to a pin, the end of the wire is considered to be fixed. To disconnect a wire from a pin, turn off the **Keep Wires Connected to** option. This will release the end of the wire and let you disconnect from the pin while stretching the wire.

When a segment that is attached to a non-wire object such as a pin, and the pin is not being stretched, a segment is added from the end of the stretched segment back to the original connection point in the pin only if it is necessary to maintain connectivity.

Stretch: Keep Connected: Spine Wires Only

Example of **Keep Wires Connected to**: Choose **Spine wires only**.



Only the Spine wires will stretch.

Stretch Form

- **Lock Angles** prevents you from changing the angle of a corner or edge as you stretch it.
- **Snap Mode** controls the direction in which you can stretch an edge. Environment variable: *snapMode*
- **Delta X,Y** lets you specify the number of user units the object will stretch on the X and/or Y axes relative to its point of origin.
- **Apply** stretches the origin point of the selected object as specified in the Delta X and Delta Y fields.
- **Connections: Keep Wires Connected to** stretches paths, pathsegs, and via vertices coincident with a selected vertex. These are stretched as a set.
- **All wires and vias** maintains pathseg-to-pathseg and pathseg-to-via connections during stretch.
- **Spine wires only** maintains connections to spine pathsegs but not to connecting vias when stretching a spine. This option is selected by default. For information about selecting a spine, see *Selecting a Path Vertex*.
- **No wires and vias** does not maintain pathseg-to-pathseg and pathseg-to-via connections during stretch.
- **Keep wires connected to shapes**, if selected, maintains connection with shapes, such as pin figures, rectangles, polygons, and MPP, coincident with the selected vertex during the stretch operation. This checkbox is selected by default. Environment variable: *keepWiresConnectedToNonWires*
- **Keep guard ring ends connected**, if selected, enables you to keep the two touching ends of a guard ring together while stretching the guard ring. If the checkbox is not selected, only the selected end of the guard ring is stretched and results in opening up the guard ring. This checkbox is not selected by default.
- **Chain Mode** lets you move device chains. (You can only completely select instances, not parts of instances, so that the stretch command will only allow you to move them using the stretch command. To stretch parts of instances, you must use the stretch handles.) The options available are *All*, *Selected*, *Selected Plus Left*, and *Selected Plus Right*.
- **Special Objects: Via Mode** defines the mode of stretching of the via. The options available are *Stretch Row/Col* and *Stretch Metal*. Environment variable: *stretchViaMode*
- **Allow opposite metal extend** ensures that the cut box stays at the same location even if the enclosure is increased after the stretch operation. The metal of the new via goes beyond the existing metal. This option is available only when the *Stretch Row/Col* via mode is selected.

Module Summary

In this module, you learned how to

- Generate shapes using Layer Generation Booleans
- Stretch objects using various connected and spine options



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Labs

Lab 3-1 Sizing and Layer Generation

Lab 3-2 Using the Stretch Options



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Quiz: Creating a Layer Using Boolean Functions



Which menu command lets you use Boolean functions to create a new layer? Select all that apply.

- A. MarkNet
- B. Layer Generation
- C. Remaster Instances
- D. Find/Replace
- E. None of the above

Answer: B

The **Layer Generation** menu command lets you use Boolean functions to create a new layer.



Answer: B

The **Layer Generation** menu command lets you use Boolean functions to create a new layer.

Quiz: The Stretch Command



You can use the **Stretch** command to stretch diagonal edges, intersecting wires, and pathSegs or paths by changing the vertices of the XY locations.

- A. True
- B. False

Answer: A

You can use the **Stretch** command to stretch diagonal edges, intersecting wires, and pathSegs or paths by changing the vertices of the XY locations.



Answer: A

You can use the **Stretch** command to stretch diagonal edges, intersecting wires, and pathSegs or paths by changing the vertices of the XY locations.

Quiz: The Boolean Functions in the Layer Generation Form



What are the Boolean functions available in the **Layer Generation** form?

The Boolean functions available in the **Layer Generation** form are *AND*, *AND NOT*, *OR*, *XOR*, *GROW BY*, and *GROW BY DIRECTION*.



What are the Boolean functions available in the **Layer Generation** form?

The Boolean functions available in the **Layer Generation** form are *AND*, *AND NOT*, *OR*, *XOR*, *GROW BY*, and *GROW BY DIRECTION*.

Quiz: The Stretch with Keep Wires Connected to Options



What are all the **Stretch with Keep Wires Connected to** options?

Spine wires only, All wires and vias, and No wires and vias are the **Stretch with Keep Wires Connected to** options.



What are all the **Stretch with Keep Wires Connected to** options?

Spine wires only, All wires and vias, and No wires and vias are the **Stretch with Keep Wires Connected to** options.



Module 4

Hierarchical Editing

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Module Objectives

In this module, you

- Use the Search and Navigator Assistants to find and identify components
- Save and restore a selected set of objects to retrieve it later
- Work with Groups to filter and manage design objects
- Find and replace objects
- Remaster instances on your design hierarchy



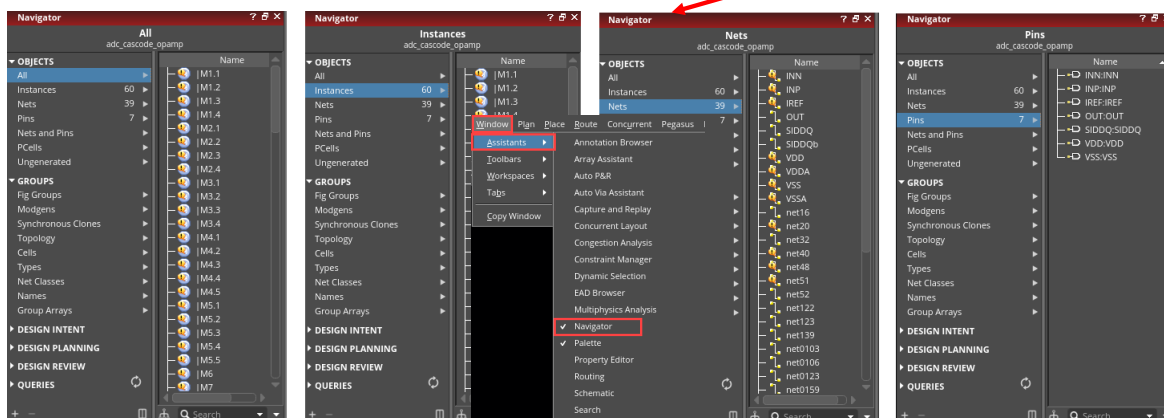
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Navigator Assistant Setup



In the layout canvas, choose **Window – Assistants – Navigator** to access the Navigator assistant.

The *Navigator* is a dockable/undockable assistant pane that provides facilities to view objects across the design hierarchy using a tree representation.



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Navigator

You access the Navigator assistant by selecting the **Window – Assistants – Navigator** menu entry.

The Navigator assistant is a dockable assistant pane that provides facilities to view objects across the design hierarchy using a tree representation.

The Navigator Tree is a hierarchical expandable/collapsible navigation system, used to display devices, nets, and pins that exist in the design area. It helps you navigate through a hierarchical design to find the components you are looking for. Selecting a component in the Navigator selects the corresponding device in the design area. Similarly, selecting a component in the design areas selects the component in Navigator.

If you select a net in the Navigator assistant, the physical shapes on the net are highlighted/probed in the design canvas. Note that the shapes are only highlighted/probed but not selected. When you delete a net selected using the Navigator assistant, it deletes the associated shapes on the net. The pins on the net are not deleted.

If you select a pin or an instance in the Navigator assistant, the design display area automatically pans to or zooms at the selected object if not currently visible. You can control whether to pan or zoom to the selected object using the `autoZoomMode` environment variable. You can turn off the automatic panning or zooming by setting the `autoZoomPan` environment variable to **nil**.

Note: In the Navigator assistant, all nets that connect to an instance within the group are displayed as duplicate nets inside that group.

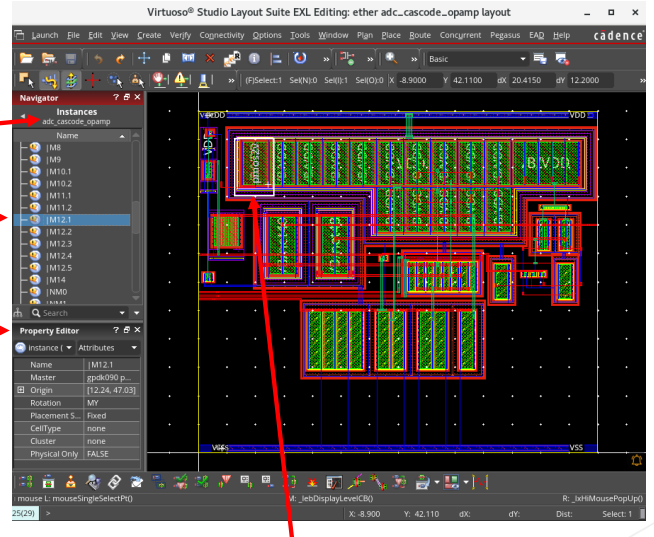
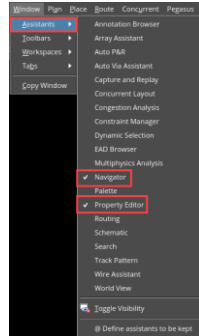
For detailed information about the Navigator user interface and how to work with the Navigator, see “The Navigator Assistant” in the *Virtuoso Schematic Editor XL User Guide, Product Version IC25.1*.

Navigator and Property Editor Assistants



In the layout canvas, choose **Window – Assistants – Property Editor** to access the Property Editor assistant.

If the **OBJECTS** category is set to **Instances** in the Navigator assistant, selecting an instance by name in the Navigator will show its properties in the Property Editor assistant AND highlight it in the design canvas.



Instance will highlight in the design canvas.

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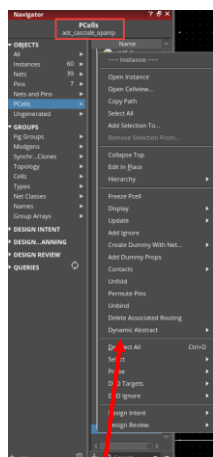
You can use the Navigator to select instances, nets, pins etc. The Navigator also lists the selected elements as you select them in the design. As you select elements in the design, the Navigator also highlights them and displays their properties in the Property Editor.

Navigator supports two major categories, **OBJECTS** and **GROUPS**, and further detail views.

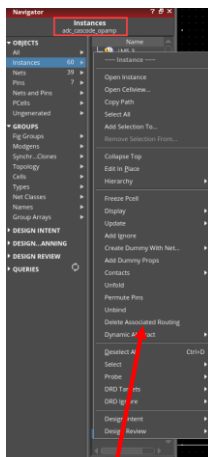
- The **OBJECTS** category has All, Instances, Nets, Pins, Nets and Pins, PCells, and Ungenerated – shows object counts upfront.
- The **GROUPS** category has Fig Groups, Modgens, Synchronous Clones, Net Classes etc.
- *Current* and *Below* filters are replaced with the *Switch hierarchy display* button.
- The *Toggle “Side By Side” display* button is used to arrange the sets and their contents sideways.

Selection and toggle settings are saved in the `./cadence/dfII/Navigator/Options.xml` file.

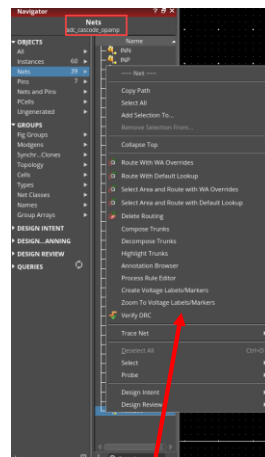
Right Mouse Button Navigator Options



Options for PCells



Options for Instances



Options for Nets

There are specific options for each **Navigator OBJECTS** category.

The Navigator assistant can also be used to perform operations on the selected objects.

Using the right mouse button on one of the entries displayed in the Navigator will show a contextual menu listing all the operations available.

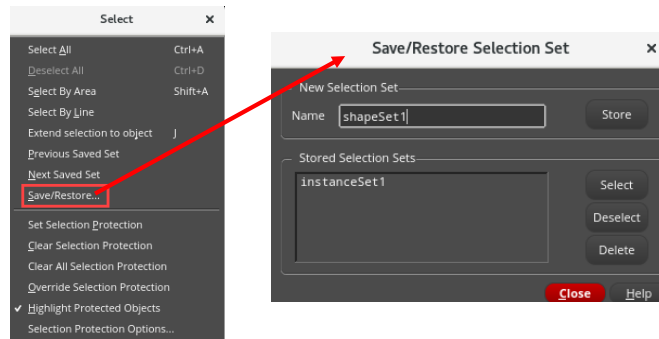
Note: The Navigator assistant is very helpful to select “instances” or “pins” only. Remember that everything that is selected in the Navigator is selected in the layout view as well.

Invoking the Save/Restore Selection Set Form



In the layout canvas, choose **Edit – Select – Save/Restore** to invoke the Save/Restore Selection Set form.

The Save/Restore Selection Set form lets you save and restore a selection set (full or partial). You name each set for subsequent recall.



Save/Restore Selection Set

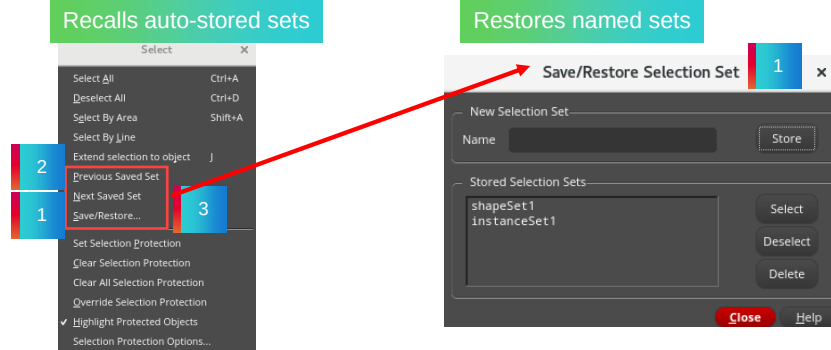
- **Name** allows you to enter a name to a selected set of objects in order for you to retrieve it later.
- **Store** allows you to save this information to a file for later retrieval. If the Store button does not work, the selection set is empty, or the name you are entering in the name field already exists. A warning is issued in the CIW.
- **Stored Selection Sets** are entered into the field once you click the **Store** button.
- **Select** retrieves the stored information and displays it on the originally selected set. You must click the stored selection set before clicking the **Select** button to bring up the stored selection set you want to display. Each selection you make adds an additional selection to the display unless you click the **Deselect** button from the form before making another selection.
- **Deselect** removes the selection from the stored set from the objects in the stored selection set you have selected to display in the cellview.
- **Delete** removes the set you have selected in the Stored Selection Sets form window. If the deleted set is the current selection set, the selection is removed from the form but remains displayed until you change the selection.

Recalling the Selected Sets



You can recall the auto-stored sets or restore the named sets.

1. You can iterate through the sets by selecting the sets in the **Save/Restore Selection Set** form.
2. You can iterate through the sets by choosing the **Edit – Select – Previous Saved Set** command.
3. You can iterate through the sets by choosing the **Edit – Select – Next Saved Set** command.



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You can recall the auto-stored sets or restore the named sets.

You can iterate through the sets by selecting the sets in the **Save/Restore Selection Set** form or using the **Edit – Select – Previous Saved Set** or **Edit – Select – Next Saved Set** command.

When using Previous Saved Set, the last selected set is again selected. When you click on Previous Saved Set again, it moves back to the selected set, which was active prior to the last state.

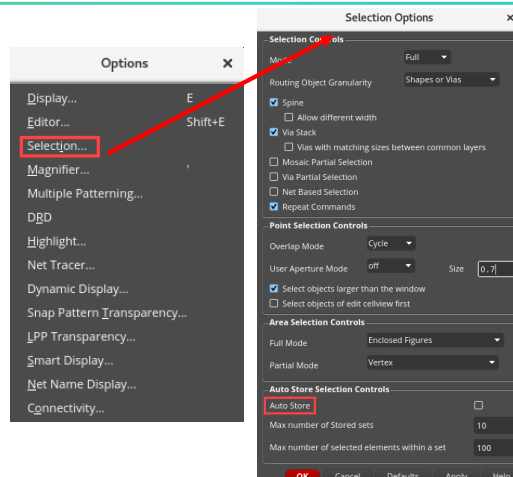
The **Save/Restore** command brings up the **Save/Restore Selection Set** form from which you can choose from a list of saved sets.

Using the Auto Store Selection Controls



In the layout canvas, choose **Options – Selection** to open the Selection Options form.

- The **Auto Store** option will let you restore your previous selections without having to explicitly save them.
- This can be useful, for instance, to prevent accidental losses of your selected set:
 - The layout designer has selected several objects.
 - The designer then wants to add another object but forgets to hold the **Shift** button.
 - Only the last object will be selected, and the previous selection work is lost.
- In this case, it is enough to recover the previous selection using the **Previous Saved Set** menu entry.



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The **Selection Options** form comprises the following sections, namely **Selection Controls**, **Point Selection Controls**, **Area Selection Controls**, and **Auto Store Selection Controls**.

The **Auto Store Selection Controls** section stores multiple sets of selections you have made. You can reload these selections into the cellview without having to reselect the objects. The radio button controls whether or not you can store the selections (environment variable: *autoStoreSelectionSetOn*).

- The *Auto Store* option is disabled by default.
- The selection set consists of full or partially selected figures.
- The selection set is a cellview property. **Edit – Select – Save/Restore** is not cumulative. Once the cellview is closed, the recorded information is not retained.
- You can iterate through the sets by selecting the sets in the Save/Restore Selection Set form or using the **Edit – Select – Next Saved Set** or **Edit – Select – Previous Saved Set** command.
- Deleted or modified objects are removed from the saved selection set. If all objects are removed from the selection set, the set is removed from the Save/Restore Selection Set form.
- Restore is independent of the object selectability of the Layer Selection Window and the Object Selection Window.
- The command is not compatible with the Undo or Redo command.
- If the subview is already open, the command does not work with Edit In Place or the Descend command.

Max number of Stored sets limits the number of sets that can be retained by the Save/Restore Selection Set form (environment variable: *maxAutoStoreSelectionSet*).

Max number of selected elements within a set limits the number of objects you can enter into one selection set (environment variable: *maxAutoStoreSelectionObj*).

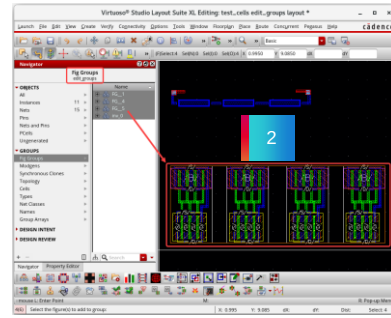
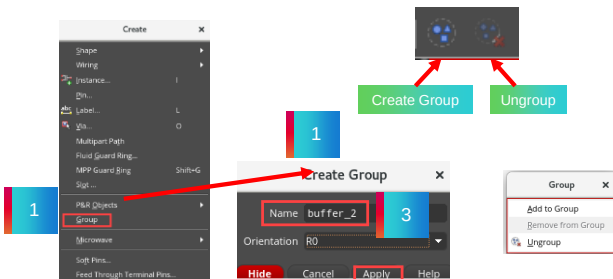
Creation of Groups



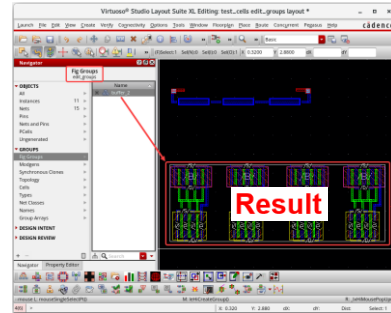
The **Group** command lets you group shapes and/or instances together for you to manipulate the objects as a whole, rather than individually.

1. Choose **Create – Group**.
2. Select the objects that should form the group.
3. Provide the name of the group and select **Apply**.

Result: The objects are now bound in a group.



Individual shapes



Single Group

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The Group command allows you to group shapes and/or instances together in order for you to manipulate the objects together, rather than individually. It will have as an owner an area boundary object, which will conform to the bounding box of the entire group (the composite of all of the bounding boxes of the members of the group). This area boundary is automatically created, edited, and deleted by the Group command as objects are added to or deleted from the group.

A group:

- Is a named database figure, which you can query in the database.
- Can contain any number of figures or instances, including other groups.
- Is selectable.
- Has an origin and an orientation.
- Does not have a layer/purpose, though its members may have.
- Does not have connectivity, though its members may have.
- May have an applied constraint group.

After a group is created, it can be edited using the commands listed below:

- Edit – Group – Add to Group:** Select an object which is not included in a group yet, and add it to one of the existing groups.
- Edit – Group – Remove from Group:** Assuming the currently selected object is in a group, with this command, it will be removed from it.
- Edit – Group – Ungroup:** Destroys the currently selected group. Of course, the objects in the groups will not be deleted.

The Create Group and Ungroup commands have the corresponding toolbar buttons in the Edit toolbar.

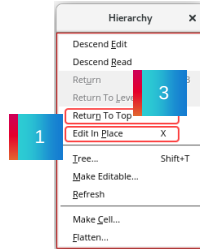
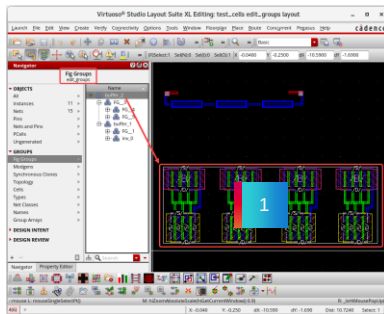
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Groups: Creating and Editing the Content



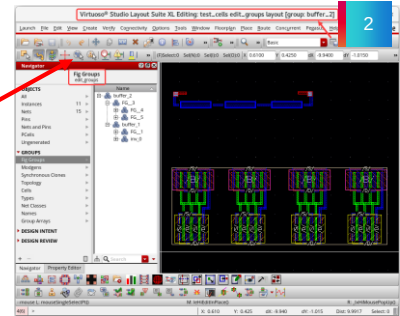
To edit the objects included in a group, the same approach used for hierarchical instances with **Edit In Place** command has to be used.

1. Select the group to edit and “descend” using **double-click** or the **Edit In Place** (bindkey X) option.
2. The name of the group will be displayed in the window title.
3. When the edits are completed, return to the “top” level using the **Return To Top** command.



Transparent Group

The **Transparent Group** option is used to temporarily flatten all the groups in the current view.



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You can edit the contents of a group either by using the **Edit In Place** command or by switching over to the transparent group mode.

To edit the objects included in a group, the same approach used for hierarchical instances has to be used.

Using Edit In Place

1. Click the group whose contents you want to edit.
2. Choose **Edit – Hierarchy – Edit In Place** (bindkey X). Alternatively, you can **right-click** the group to display the Group context-sensitive menu and select **Edit In Place**.
3. Perform the desired edit operations on the objects within the group.
4. Choose **Edit – Hierarchy – Return to Level** or **right-click** anywhere in the design canvas and select **Return** from the Layout context menu.

The name of the group is reported in the window title, in a similar fashion, to the name of the sub-cells during an Edit In Place operation.

The **Transparent Group** option can be used to temporarily “flatten” all the groups in the current cellview.

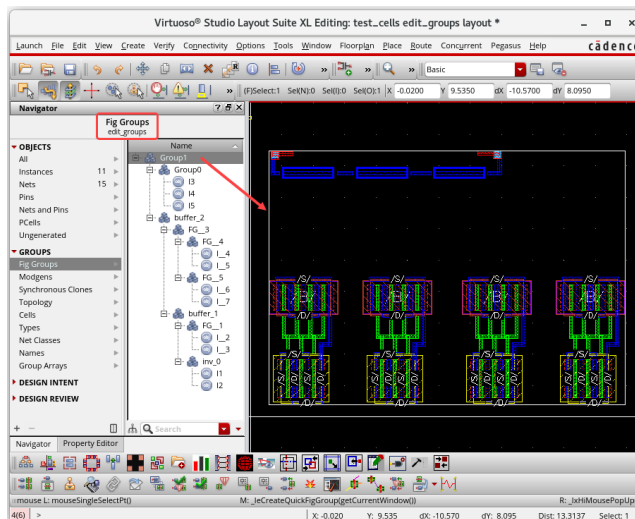
After having performed the required edits, the groups can be “restored” by simply deselecting this option.

This feature is very useful in case it is needed to align or to move in one shot several objects included in different groups.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC23.1*, for more details on these features.

Hierarchical Groups

Select existing groups and create a new group to build a hierarchical group. Any mix of groups and simple objects can be used when building groups.

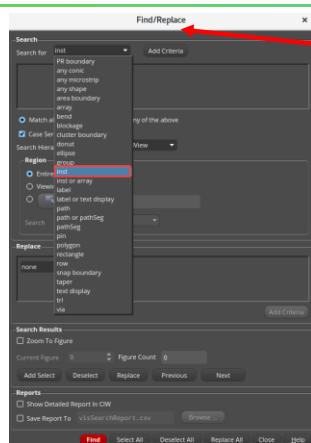


Virtuoso® Layout Suite supports hierarchical groups and the Navigator reports them in the tree view.

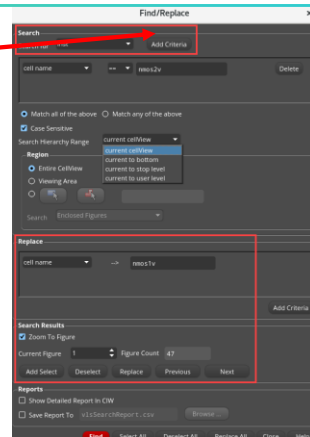
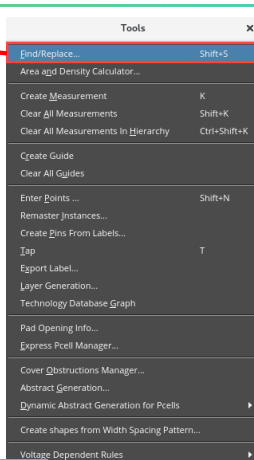
Finding and Replacing the Database



To locate a specific database object, in the layout canvas, choose **Tools – Find/Replace** or press the bindkey **Shift+S** to invoke the Find/Replace form.



Search Hierarchy Range: current cellView, current to bottom, current to stop level, and current to user level.



Search for: Set to find instances of nmos2v in the current cellView.

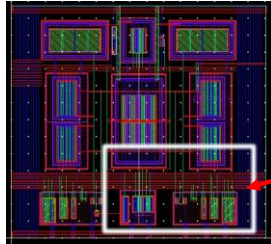
To search for data by type, use the **Find/Replace** command. You first specify the type of object you want to search for. After the search is completed and the objects are highlighted, you can specify whether you want to add one object at a time to the selected set or add all objects found in the search. Once the objects are selected, you can replace them with new values. The Find/Replace command works either in a cellview or hierarchically.

In the Find/Replace form, you search for the type of objects to search for from the pull-down list in the form. You then choose where in the hierarchy to look, i.e., *current cellView*, *current to bottom*, *current to stop level*, and *current to user level*. When you click **Find**, the system searches for all specified objects. You can then add these objects to a selected set and replace various criteria associated with them or replace criteria without selecting them.

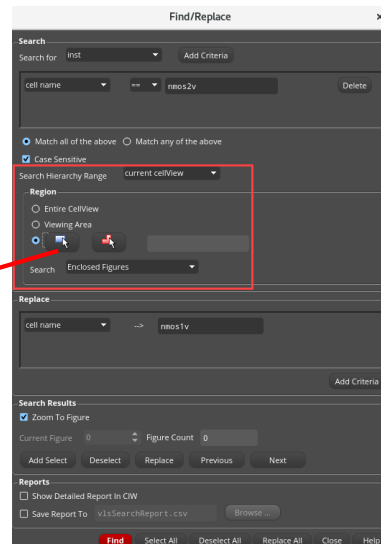
For example, you can search for all instances of the cell *nmos2v* in the *current cellView* and replace them with the cell *nmos1v*. You can also search the hierarchy from current to bottom for all paths of a certain width and change their widths. You can search for all rectangles on a specific layer, add them to the selected set, and delete them.

Find/Replace: Region Section

- The user can specify a search region in the Find/Replace form.
- The search region can be the Entire CellView, the current Viewing Area, a user-drawn rectangle region, or a polygon region.
- A smaller search region would reduce the search time and the number of search results.



Click on the **Rectangle** or the **Polygon** button to draw a rectangle or polygon region. The region will be highlighted.



The Find/Replace form comprises the following sections:

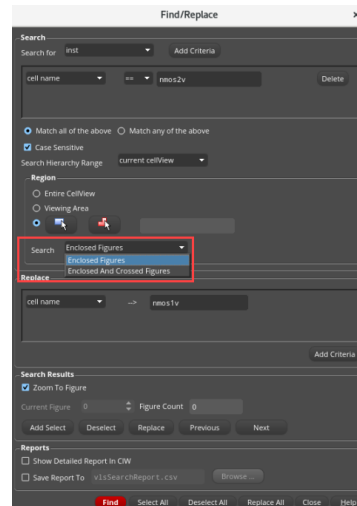
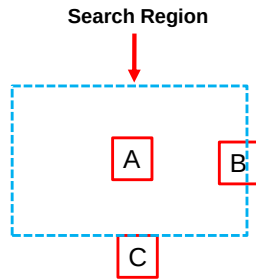
- Search
- Region
- Replace
- Search Results

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for details on these features.

Find/Replace: Search Section

- **Enclosed Figures:** The objects completely enclosed by the viewing area or the rectangle/polygon area.
- **Enclosed And Crossed Figures:** The objects completely or partially enclosed by the viewing area or the rectangle/polygon area.

Example
Enclosed Figures: A
Enclosed And Crossed Figures: A & B



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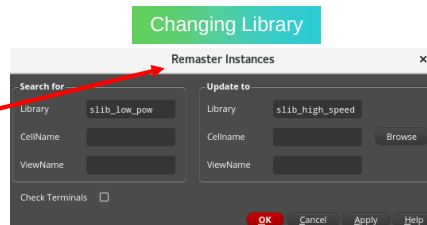
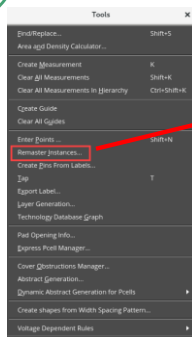
Region

- **Entire CellView** considers the entire cellview for the Search command. This option is selected by default.
- **Viewing Area** considers the portion of the cellview that is displayed in the canvas at that instant for the Search command.
- **Draw a Rectangle Region on the Canvas** or **Draw a Polygon Region on the Canvas** enables you to specify a rectangular or polygonal area for the Search command. Select the required option, and then click in the canvas to specify the vertices of the rectangle or polygon.
- **Search** sets the object searching for. The option is enabled only when the viewing area or the rectangle or polygon area in the *Region* section.
- **Enclosed Figures** searches the objects completely enclosed by the viewing area or the rectangle or polygon area.
- **Enclosed And Crossed Figures** searches the objects completely or partially enclosed by the viewing area or the rectangle or polygon area.

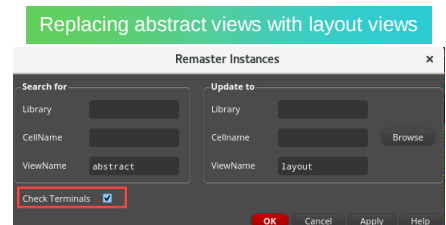
Remastering the Instances



To remaster the instances in the layout canvas, choose **Tools – Remaster Instances** to invoke the Remaster Instances form.



All instances in the current cellview that have masters from the library “slib_low_pow” are remastered to the library “slib_high_speed” and their master cell/view names are not changed.



Instances are remastered only in the current cellview. Turn on **Check Terminals** to check all corresponding cellviews for matching or a subset of matching instTerm information (optional).

The **Check Terminals** option checks instTerms of cellviews for matching or a subset of matching instTerms. When **Check Terminals** is on, and the instTerms of the cellview in the **Search for** field are a subset of the instTerms of the cellview in the **Update to** field, or they are equivalent, the cellviews will be remastered.

The **Remaster Instances** command is capable of remastering cellviews based on the view names, cell names, and the library in which they reside. For example, remastering the abstract views with layout views directly before translating the final design to the Stream format provides optimal performance. Replacing cellviews before editing the design can add unnecessary complexity to the design for routing and editing purposes.

You have a number of options for remastering cellviews.

By leaving text fields blank in the form, entire libraries and cellviews can be searched for matching cellviews and view names.

You can use the * wildcard to replace any number of characters prior to or following the asterisk.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for details on these features.

Module Summary

In this module, you learned how to

- Use the Search and Navigator Assistants to find and identify components
- Save and restore a selected set of objects to retrieve it later
- Work with Groups to filter and manage design objects
- Find and replace objects
- Remaster instances on your design hierarchy



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Labs



Lab 4-1 Using the Search, Save/Restore, and Navigator Assistant

Lab 4-2 Working with Groups

Lab 4-3 Searching the Database

Lab 4-4 Remastering Instances



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Quiz: The Assistants to Find and Identify the Components



What are the assistants can you use to find and identify the components in the layout design? Select all that apply.

- A. Search
- B. World View
- C. Navigator
- D. Palette
- E. All of the above

Answers: A and C

You can use the **Search** and **Navigator** assistants to find and identify the components in the layout design.



Answers: A and C

You can use the **Search** and **Navigator** assistants to find and identify the components in the layout design.

Quiz: The Save/Restore Selection Set Form



The **Save/Restore Selection Set** form lets you save and restore a selection set (full or partial).

- A. True
- B. False

Answer: A

Yes. The **Save/Restore Selection Set** form lets you save and restore a selection set (full or partial).



Answer: A

Yes. The **Save/Restore Selection Set** form lets you save and restore a selection set (full or partial).

Quiz: Combining the Shapes and/or Instances Together



Which command lets you combine shapes and/or instances together for you to manipulate the objects as a whole, rather than individually? Select the correct choice.

- A. Combine
- B. Group
- C. Instance
- D. Shape
- E. None of the above

Answer: B

The **Group** command lets you group shapes and/or instances together for you to manipulate the objects as a whole, rather than individually.



Answer: B

The **Group** command lets you group shapes and/or instances together for you to manipulate the objects as a whole, rather than individually.

Quiz: The Bindkey of the Find/Replace Command



What is the bindkey used to execute the **Find/Replace** command? Select the correct choice.

- A. Ctrl+S
- B. Shift+F
- C. Shift+R
- D. Shift+S
- E. None of the above

Answer: D

You can use the **Shift+S** bindkey to execute the **Find/Replace** command.



Answer: D

You can use the **Shift+S** bindkey to execute the **Find/Replace** command.

Quiz: The Remaster Instances Command



The **Remaster Instances** command is capable of remastering cellviews based on the view names, cell names, and the library in which they reside.

- A. True
- B. False

Answer: A

Yes. The **Remaster Instances** command is capable of remastering cellviews based on the view names, cell names, and the library in which they reside.



Answer: A

Yes. The **Remaster Instances** command is capable of remastering cellviews based on the view names, cell names, and the library in which they reside.

Quiz: Recalling the Selected Sets



What are the commands available in **Select** menu in the layout canvas to recall the selected sets?

The commands available in **Select** menu in the layout canvas to recall the selected sets are *Previous Saved Set*, *Next Saved Set*, and *Save/Restore*.



What are the commands available in **Select** menu in the layout canvas to recall the selected sets?

The commands available in **Select** menu in the layout canvas to recall the selected sets are *Previous Saved Set*, *Next Saved Set*, and *Save/Restore*.

Quiz: The Selection Options Form



Which option in the **Selection Options** form will let you bring back your previous selections without having to explicitly save them?

The *Auto Store* option in the **Selection Options** form will let you bring back your previous selections without having to explicitly save them.



Which option in the **Selection Options** form will let you bring back your previous selections without having to explicitly save them?

The *Auto Store* option in the **Selection Options** form will let you bring back your previous selections without having to explicitly save them.

Quiz: Editing the Objects Included in a Group



Which command can you use to edit the objects included in a group?

You can use the **Edit In Place** command to edit the objects included in a group.



Which command can you use to edit the objects included in a group?

You can use the **Edit In Place** command to edit the objects included in a group.

Quiz: Locating and Replacing the Database Object



Which command can you use to locate and replace a specific database object?

You can use the **Find/Replace** command to locate and replace a specific database object.



Which command can you use to locate and replace a specific database object?

You can use the **Find/Replace** command to locate and replace a specific database object.



Module 5

Course Conclusions

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Summary

In this course, you learned to

- Use the features of the Create Via command
- Automatically generate labels with existing information using the Create Label command
- Use the various options of the Copy and Repeat Copy commands
- Generate shapes using Layer Generation Booleans
- Stretch objects using various connected and spine options
- Use the Search and Navigator Assistants to find and identify components
- Save and Restore a selected set of objects to retrieve it later
- Work with Groups to filter and manage design objects
- Find and replace objects
- Remaster instances on your design hierarchy



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References

Refer to the links below for further details on the topics in this course.

[Virtuoso Layout Suite XL: Basic Editing User Guide Product version IC25.1](#)



[Virtuoso Layout Suite XL: Basic Editing User Guide Product version IC25.1](#)



Module 6

Next Steps

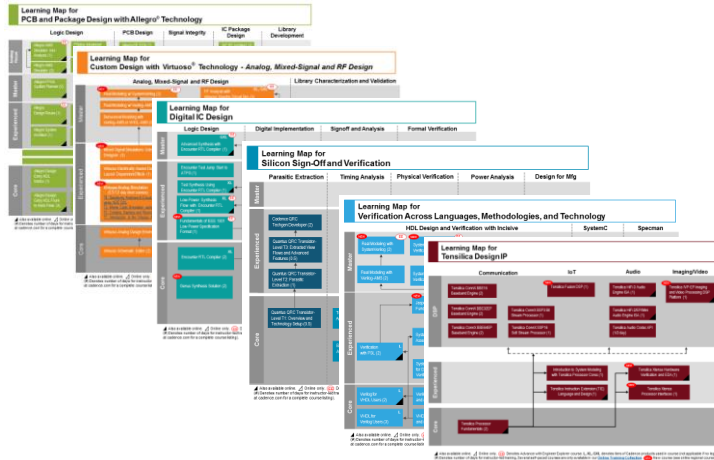
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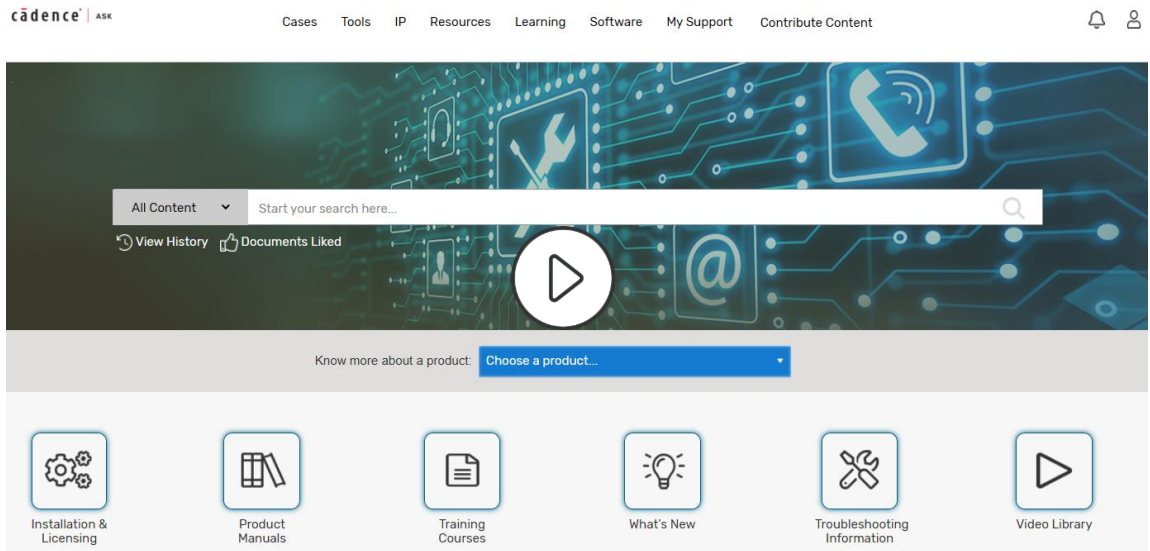
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http://www.cadence.com/Training/Pages/learning_maps.aspx

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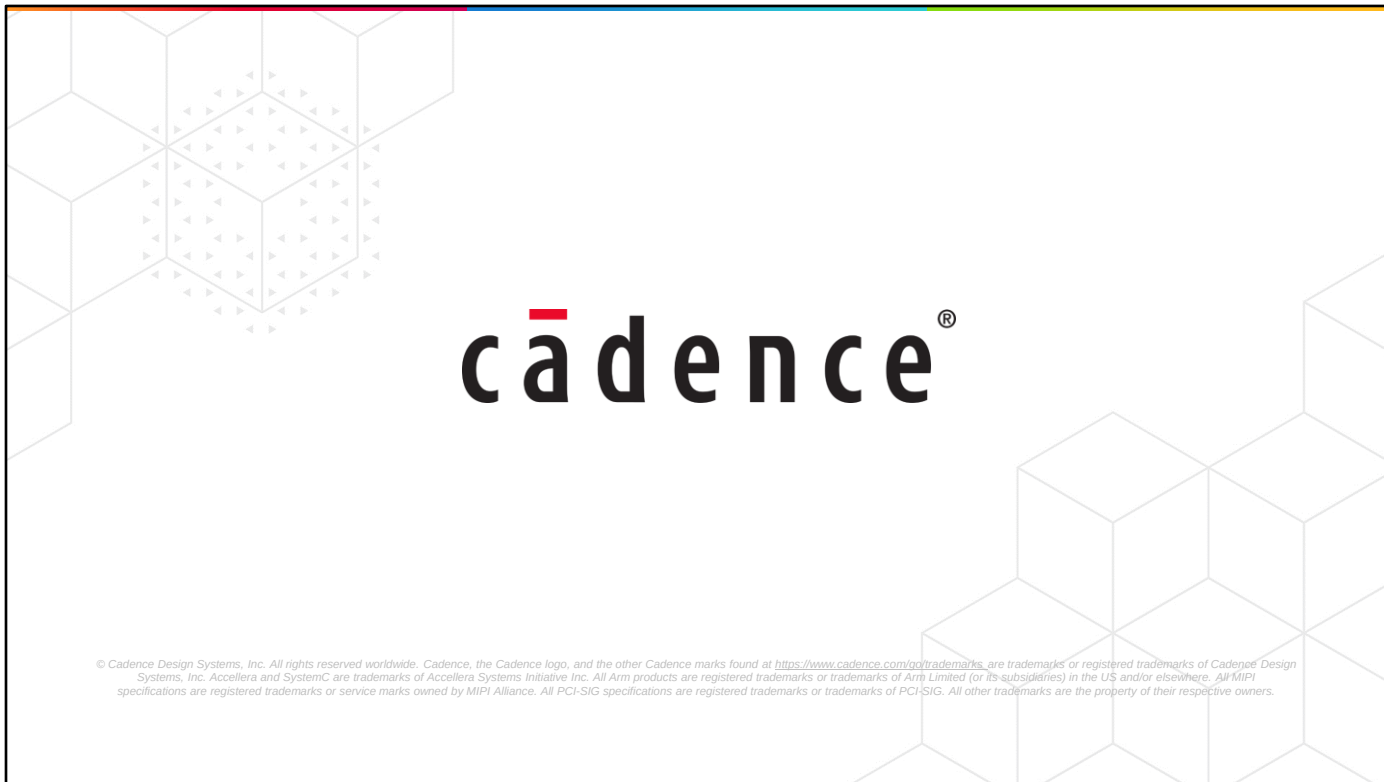
Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion

Thank you!



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